

Day 6 — SAP CPI Components (25 Aug 2026)

1. Recap — CPI as a Middle System

CPI → a middle system which is helpful in integrating two systems.

The four combinations CPI must handle:

SAP	→	SAP
SAP	→	non-SAP
non-SAP	→	SAP
non-SAP	→	non-SAP

2. The "Router/Wire" Analogy

Your trainer used a simple, everyday analogy: **let's say we want to connect two systems** — think of it like a **Router** connected to a **System** through a **wire**, and that same router also connects out to **CPI**.

The idea being illustrated: Just like a router is the physical device that sits in between and lets two devices on a network talk to each other, CPI plays that same "in-between" role for two software systems — it's the connector that lets otherwise-separate systems exchange data.

3. What CPI Actually Does — Source to Target

CPI → picks the data from the Source and transfers that data to the Receiver System.



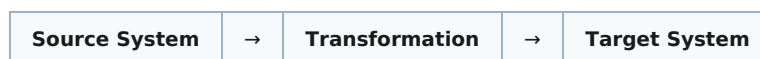
The data flowing through this chain can be referred to by three interchangeable terms: **Data / Message / Information**.

4. Security While Transferring Data

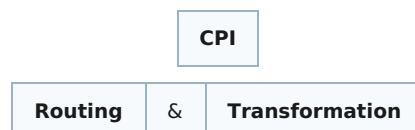
While transferring the data, that data should be secure. This is a core principle — any data moving between Source → CPI → Target must be protected (encryption in transit, proper authentication, etc.), not sent in the open.

Terminology clarified: Information, as it moves through the flow, is generally referred to as a **Message** (i.e., Data → becomes a Message once it's in motion through CPI).

5. The Three-Part Flow: Source, Transformation, Target



CPI's two core jobs, shown as a simple diagram in your notes:



What Each One Means

Function	Definition
Routing	Routing the data to different systems

Transformation	We will modify the existing data and transfer it to the Receiver System in the required format
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6. Discover Tab — Creating an iFlow from Existing Flows

Your trainer explained the **Discover** tab and demonstrated **creating an iFlow using existing flows under the Product tab** — meaning you don't always start from a blank canvas; you can copy/base your iFlow on content SAP has already built for a specific product.

Example Fields Discussed

Order, Employee, Address, Billing Information — examples of common data categories you'd see represented in these predefined flows.

7. Worked Example — S/4HANA to Commerce Cloud

S4/HANA	Message →	CPI	Message →	Commerce Cloud
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8. Effort Estimation — Why Predefined iFlows Matter

This is a genuinely important real-world point from your notes — a classic project estimation scenario:

Scenario	Estimate
1 Interface (built from scratch)	~5 days
100 Interfaces (built from scratch)	Would be extremely time-consuming if done one by one (this is the "?" your notes leave open — the point being made is that this doesn't scale)

Why this matters: If SAP → common interfaces already exist as **predefined iFlows** that **all companies** can reuse (since many companies need near-identical interfaces — this connects back to the FedEx/DHL/UPS example from Day 5), the effort for those 100 interfaces drops dramatically compared to building every single one manually. This is the real business case for why the Discover tab and predefined content exist — it's a genuine project time-and-cost saver, not just a learning convenience.

9. Standard Content — Brief Overview

Standard Content = Predefined tabs/packages. This is SAP's library of ready-made integration content.

10. Design Tab — Where Custom Flows Are Built

- **Custom flows → Design tab**
- **The Design tab is the one and only tab we will use — nowhere else in SAP CPI** is where you build your own custom integration content

Clarification needed: Your notes mention "**PO system explanation — 2 places**" — this phrase is unclear/likely a partial note from class (possibly referring to Purchase Order as an example data object discussed in two different contexts, or it could be a mishearing of something else your trainer said). If you remember more context on this point, share it and I'll fold in an accurate explanation — I don't want to guess and give you inaccurate information here.

11. About Package — Recap and Expansion

- **Package → iFlows → step-by-step process** — this is what we (as developers) actually build
- **We are responsible for creating the custom iFlows**
- **All the group of iFlows → Package** (i.e., a Package is simply the container/grouping for a set of iFlows)

Discover Tab vs. Design Tab — The Key Difference

Tab	Analogy Used in Class	What It Means
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Discover Tab	<i>Example: Google Drive</i>	There's only one Discover tab — think of it like a shared drive of ready-made content anyone can access, but not directly edit in place
Design Tab		Where you actually build and own your custom content

Important rule: If we want to use any iFlow that exists in the Discover tab, we need to **Copy** it — nothing else we can do directly in the Discover tab (i.e., you can't edit SAP's predefined content in place; you copy it into your own Design tab workspace first, then customize your copy).

Pre-Existing vs. Post-Existing Package Usage

Your notes mention: **Pre-Exist, Post-Exist package/function usage** — this refers to the general pattern of using content that already exists (pre-built by SAP) versus content you build fresh yourself (post/after, i.e., your own custom creation).

12. Now About the Monitor Tab

Message → Problem → Where you will check the failures. This is the core purpose of the Monitor tab: it's where you go when something goes wrong to investigate.

Track / Monitor Tab

Track(or) Monitor tab → will be happened in Monitor tab. Meaning: tracking a message's journey and status is a function that happens specifically within the Monitor tab.

Status of a Message

Status of Message	→	Success
	→	Failure

(Both outcomes are tracked under Monitor.)

Worked Example — Email Sending

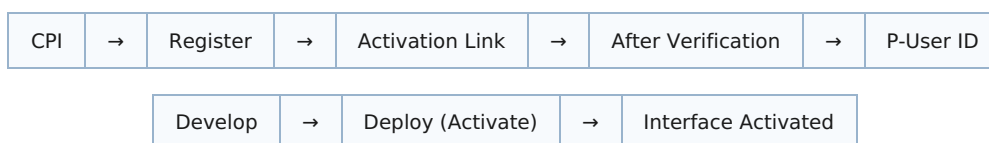
Using an email-sending scenario as the example, message statuses can include:

Status	Meaning
Message	Currently processing / in progress
Failed	Processing did not complete successfully
Retry	System is attempting to reprocess
Complete	Successfully finished

Messages Status Overview Tab — note that the terms **Interface** and **iFlow** are used interchangeably (same thing, different names depending on context).

13. Manage Integration Content Tab — The Full Lifecycle

Your notes trace the complete journey of getting an iFlow live, step by step:



So: Interface (Activate) → messages pass from one system to another system.

So: iFlow → deployed → Monitor & Manage Integration Content → Started

Summary in plain terms: Once you register with SAP and verify your account (getting your P-user), you develop your iFlow in the Design tab, then **deploy/activate** it. Once activated, it becomes a live "Interface" capable of passing real messages between systems — and from that point, its ongoing status/health is tracked under **Monitor**, while its content/lifecycle is managed under

14. Manage Security Tab

The Manage Security tab covers: **Security Material, Keystore, PGP Keys, Access Policies, JDBC Material, User Roles, Connectivity Test.**

Verified accurate — confirmed against SAP's official documentation. Quick definitions for each:

- **Security Material** — credentials (user credentials, OAuth2 client credentials, etc.) used by iFlows to authenticate to external systems
- **Keystore** — key pairs and certificates used for both inbound and outbound secure connections
- **PGP Keys** — keys required for scenarios involving digital encryption/signing using PGP
- **Access Policies** — rules restricting access to specific integration artifacts and their associated data (useful when multiple developers share one tenant)
- **JDBC Material** — connection details for scenarios using the JDBC adapter to talk to a remote database
- **User Roles** — roles restricting which sender systems can access specific iFlow endpoints
- **Connectivity Test** — tools to test outbound connections from CPI to receiver systems before going live

15. Manage Stores

Five categories listed in your notes:

1. **Data Stores**
2. **Variables**
3. **Message Queue**
4. **Number Ranges**
5. **Partner Directory**

Verified accurate. Quick context: Data Stores, Variables, and Message Queues (JMS Queues) are all storage options used when an iFlow needs to persist data mid-processing — for example, in asynchronous patterns where message content is needed again in a later processing step. Each has different use-case guidance (e.g., Data Store for larger persisted payloads, Variables for smaller reusable values, JMS Queues for asynchronous message-passing between flow steps).

16. Access Logs

Log Type	Purpose
Network Logs	Access logs and system log files
CPI Messages → Issues → Trace Logs	Used specifically to trace/debug issues within CPI message processing

17. Manage Locks

Design-time Artifact Locks — example given: **locked by username.**

What this means in practice: When one developer is actively editing an iFlow, CPI locks that artifact so a second developer can't simultaneously make conflicting changes — the lock shows which username currently holds it, so the team knows who to check with before editing.

18. Usage Details Tab

Change & Usage details.

19. Subaccount — Usage Analytics

Subaccount → Usage Analytics — tracking consumption/usage at the subaccount level, tying back to the billing/entitlement concepts covered in earlier days.